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Digital image forensics using EXIF data of digital evidence

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***Abstract*— Pie Information Extractor is one of the applications to data mining from digital photos. The global position system (GPS) with suitable connection with digital technologies enables to put additional layer to data. Exif (Exchangeable image file) information of photographs contains various information, such as date and time the picture was taken, type of camera used, types of settings on the camera, GPS Coordinates where the picture was taken, etc. Today not only digital cameras but also various devices with camera technology could put geotagged layers to photos. This information could be linked with location, camera model name, revised software, created date. Cyberattacks used not only the image layer but EXIF data are very useful. For example, the time and date of geotagged information are potentially useful for thieves and also criminologists on the other side. More and more photos are often shared on social media without any caution about private information. This paper presents another approach of image visual recognition in the interconnection with geotagged information.**

I. INTRODUCTION

Digital computer forensics is important in various areas of our everyday lives. Corporations, investigators, IT professionals, criminologists but on the other side also thieves, cyberterrorists usually use suitable methods to gain private data about the victim. In general, forensic process consists of:

- identification – identification of relevant sources of digital evidence and complete documentation of evidence material;
- collection – an important phase of digital investigation, to ensure no modification of evidence material and the verification process consist of validation information by MD5/SHA1 hash values;
- analysis – identify tools and various techniques to use and this process depends on the requirements of the investigators; its most important phase of digital computer forensic;
- presentation – report and present the evidence findings in various ways (written re-port, graphical presentation, testimony, survey).

Forensics researchers solve many obstacles with the development of new devices and software solutions. These changes need especially DFRWS organization rapid review and dissemination of up-to-date results in any of forensics area. [1] More and more products passed from printed form to digital, especially in the covid19 phase of the human being. So various products exist in a digital way such as e-books, photos, money, e-commerce, e-sports, video tutorials, distance learning, any kind of courses, e-shops, etc. All these products developed their own structure in many ways but specially to collect as much data from users and then arises our own digital footprint (in compare with human footprint it's a trail of any moving human being made on the internet). With the covid19 restrictions in many countries, if people wanted to be with the others, they have to use any of online services in many ways for free and any kind of information was recorded, shared on the internet. IT professionals often talk about the services for free – the commodity of payment are our own data (our geotagged position, time on the internet, our webcams environment, e-mails, every online page we click, face detection, and more and more photos about our private life, etc.). [2, 3, 4, 5]

Digital image forensics (DIF) mainly focuses on recovering and analyzing digital evidence in a criminal investigation process. DIF has been primarily used to focus on two problems and their solutions: the identification of the source of an image and image forgery detection. Identifying the origin of a digital image deals with the digital forensics analysis to recognize aspects, for example, the model of the camera responsible for generating the image (fig. 1). To verify the integrity of a digital image involves evaluating the contents of the file to determine if it has undergone one or more of adulteration processes to produce a forged image. [6, 7]

Figure 1. Blind digital image forensics technique [7]

II. THE GLOBAL POSITION SYSTEM AND GEOTAGGED PHOTOS

The Global Positioning System (GPS) is the satellite-based radio-navigation system owned by the United States government (USG) and operated by the United States Space Force. [8] It is provided positioning, navigation, and timing services to a GPS receiver on the Earth. As the system is owned and maintained by USG is freely accessible to anyone with a GPS receiver. These days are any GPS units widely integrated often as a micro-chip in many devices such as cars, planes, boats, drones, smartphones, digital cameras, GPS navigations, etc., or as a GPS tracker used without limitation. Much of information, often private is sharing on the earth. On the other side, more and more IoT devices connected to the

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Internet and collecting data from their sensors, which one of them could be a GPS tracker. [9]

Geotagged photos are broadly widened because of easy creation by any modern device, such as a digital camera or smartphone with a GPS device. The metadata of the photo usually consists of altitude ref., altitude, data stamp, latitude ref., latitude, longitude ref., longitude, timestamp, version ID as in figure 2 and figure 3. The information about place and date/time information are the main services o GPS. Geotagging (or geocoding) is the process of adding geographical identification metadata to resources (websites, RSS feed, images, or videos). [10] Another study explores four primary categories:

- user-related information – photo and user ID applicable to analyze the portrait of people;
- time and date information – time and date of photo, duration on the visited place;
- geographic information – latitude and longitude of the place, analysis of the spatial distribution characteristics of tourist destinations;
- text information – the most suitable tag to describe the place destination based on the analysis of the title, description, and label. [11, 12, 13, 14]

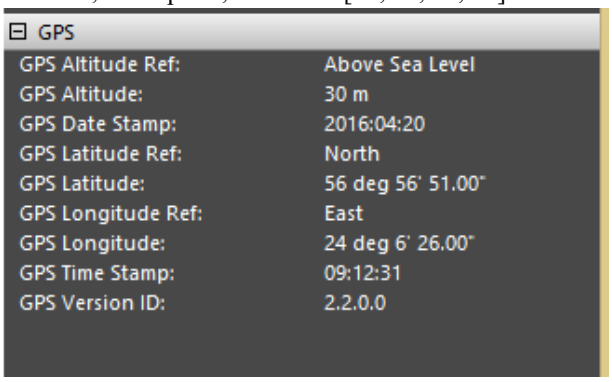

Figure 2. Capture GPS metadata of the photo

Figure 3. Visual content of the photo.

This information could be useful to identify the place and time/date of creation automatically with a suitable tool – Pie Information Extractor (PIE) as in figure 4 and figure 5 with the comparison of image.google.com. As we saw google tool is more accurate in the place and with other services offer to buy tickets to this building, web page, phone number, and the exact address of it. But it is not the rule that google tool is usually exactly. Besides this information is beneficial for us to pre-view a street view by google. With these tools is simply easy to find the accurate place is needed to be localize. The difference between these tools is about 50m between the accuracy of photography place.

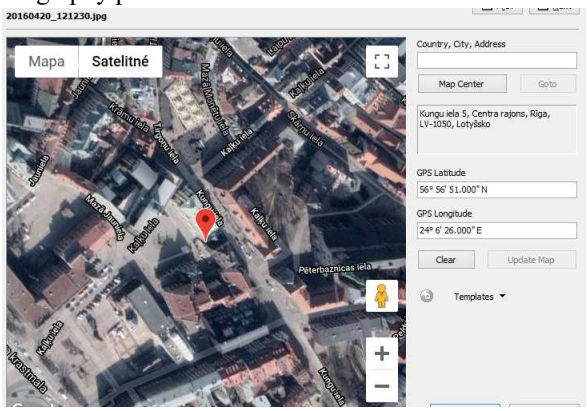

Figure 4. Identification of the exactly place on the map

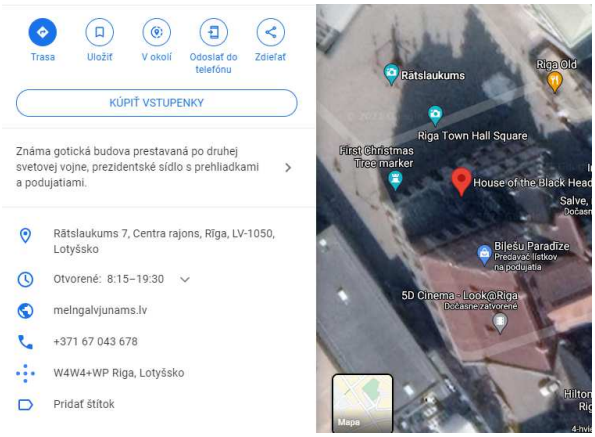

Figure 5. Google tool with the other services

This example is used when is not possible by visual content know the right place of the digital photo. With geotagged photos is no problem to categorize photos from vacations and put them on social media. The digital photo could be enhanced by GPS coordinates in various ways:

- Some digital photography devices – cameras and smartphones with built-in GPS module are able to acquire positioning, date and time information;
- GPS tracker – this device could log the position as a digital trail by the movement and put the photos accurate metadata;
- Manually add geotag to photos – user could manually put extended information to photos or sometimes the tools by sharing photos put this information by artificial intelligence. [10]

Not every digital photo has automatically positioned data, some of them are placed manually.

III. IMAGE FILE FORMAT – EXIF DATA

The first JPEG file format standard (JFIF) was defined in the year 1992 to enable the interchange of JPEG bit streams between a wide variety of applications and platforms. In the year 1998 was this technique improved to a new standard EXIF data (Exchangeable Image File Format), which allow digital camera manufacturers to put

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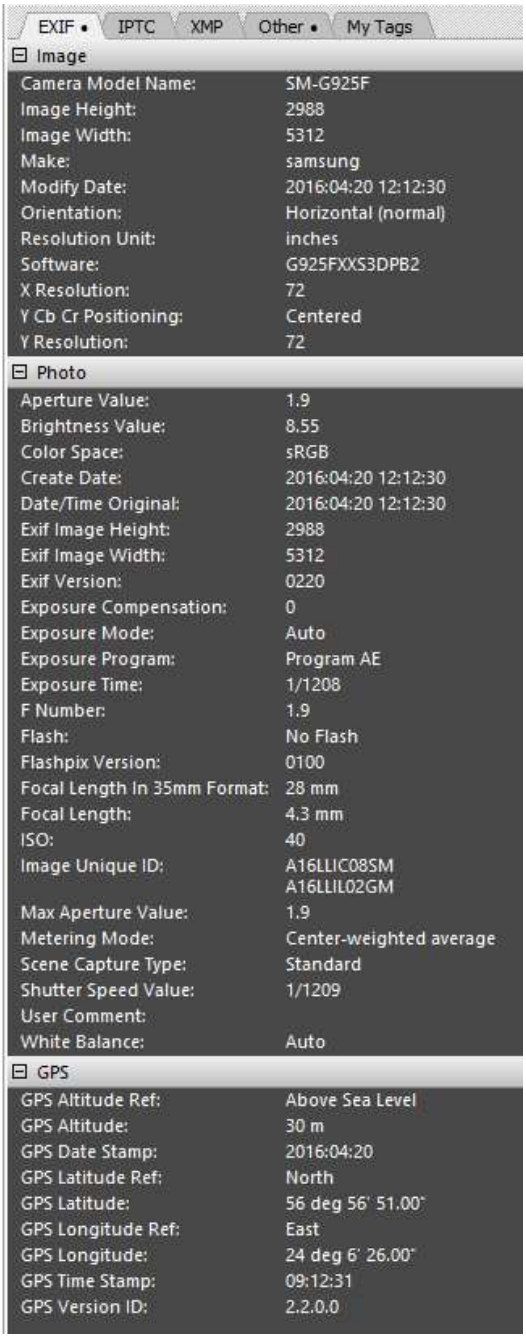
in and store camera and digital image metadata into JPEG and TIFF files. [15]

Every digital photo could be stored in various file formats in dependency of the used digital media. RAW formats contain only pixel data, while another format comprehends a wealth of information. Metadata in general provides information related to how the image file was generated and handled. This information can be used to recognize if the photo was created by a digital camera, processed by a graphical program, or altered to convey misleading information. Common properties of digital photo:

- Main information about the image – identification about the device or application that created the picture. Most digital devices include EXIF metadata block;
- Image size – the metadata often records the picture's dimensions;
- Timestamps – in general, there are information when was a photo taken;
- Types of metadata – many different types of metadata, that are generated by digital devices or by applications. The most common types of metadata include File, EXIF, IPTC, ICC Profiles, XMP, PrintIM and Photoshop. The Exchangeable Image File format (EXIF) is typically used by digital device manufacturers to identify information about the device's settings used for the photo. It typically includes timestamps, camera model, camera additional information about photosensor, lens settings, color space, exposure time, flash, and more as showed on figure 6.
- Descriptions – embedded annotations could be added to describe photo or identify the photographer;
- Missing metadata – this is often used when somebody protects own privacy. [16]

Geotagged photos are useful digital evidence as a primary or secondary source of information. Various researchers see this kind of evidence as a study of human movements and the places that have been visited. Many of people put additional information, such as keywords of event or specific place by uploading photos on web storage to making easier to find. More and more are geotagged photos used in cybercrime research as a primary source of information. [17]



EXIF • | IPTC | XMP | Other • | My Tags

Image	
Camera Model Name:	SM-G925F
Image Height:	2988
Image Width:	5312
Make:	samsung
Modify Date:	2016:04:20 12:12:30
Orientation:	Horizontal (normal)
Resolution Unit:	inches
Software:	G925FXXS3DPB2
X Resolution:	72
Y Cb Cr Positioning:	Centered
Y Resolution:	72
Photo	
Aperture Value:	1.9
Brightness Value:	8.55
Color Space:	sRGB
Create Date:	2016:04:20 12:12:30
Date/Time Original:	2016:04:20 12:12:30
Exif Image Height:	2988
Exif Image Width:	5312
Exif Version:	0220
Exposure Compensation:	0
Exposure Mode:	Auto
Exposure Program:	Program AE
Exposure Time:	1/1208
F Number:	1.9
Flash:	No Flash
Flashpix Version:	0100
Focal Length In 35mm Format:	28 mm
Focal Length:	4.3 mm
ISO:	40
Image Unique ID:	A16LLIC08SM A16LLIL02GM
Max Aperture Value:	1.9
Metering Mode:	Center-weighted average
Scene Capture Type:	Standard
Shutter Speed Value:	1/1209
User Comment:	
White Balance:	Auto
GPS	
GPS Altitude Ref:	Above Sea Level
GPS Altitude:	30 m
GPS Date Stamp:	2016:04:20
GPS Latitude Ref:	North
GPS Latitude:	56 deg 56' 51.00"
GPS Longitude Ref:	East
GPS Longitude:	24 deg 6' 26.00"
GPS Time Stamp:	09:12:31
GPS Version ID:	2.2.0.0

Figure 6. EXIF data of the photo

IV. RESULTS

This paper deals with no specific image format, but important is the data layer of the image. Digital cameras from manufacturers like Canon, Sony, or Nikon proposed to use encoding parameters as a set of features to determine the acquisition device of an image. More specifically, image dimensions, quantization, and Huffman tables were combined into a highly indicative and stable digital camera fingerprint. [18]

The application of the phases of the digital computer forensics analysis to evidence material:

- Identification - digital device recognition – the specific device model on which the digital evidence was stored;

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- Collection - digital photos validation – process of checking necessary metadata on every photo. The collection was made by 80 images robustness of metadata.
- Analysis – on the image set was performed experiments on every image visual recognition in the interconnection with geotagged information. Using only the number of used EXIF headers showed that several specific apps or online tools can be distinguished with high reliability. The analysis was made by The Sleuth Kit/Autopsy shown on figure 7, PIE shown on figure 8 and online image tool by Google;
- Presentation – report.

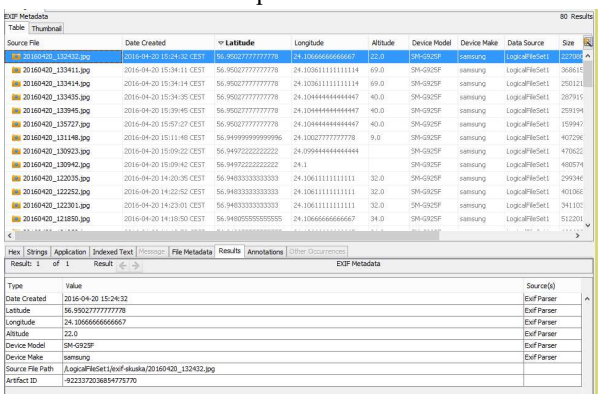

Figure 7. Image set analysis by The Sleuth Kit/Autopsy.

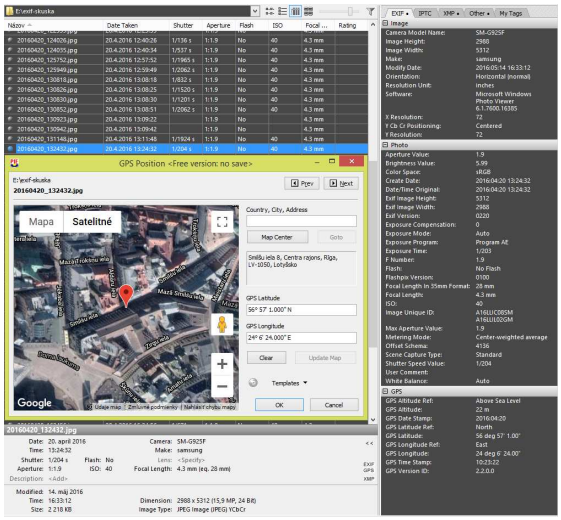

Figure 8. Image set analysis by The Sleuth Kit/Autopsy.

Compared to other approaches and automation processes by neural networks or by deep learning methods, our manual system has the benefit of standard visual content view and each file metadata descriptions analysis. With the set of 80 images is not too vast time, but on a large photo set, it could be difficult to analyze each photo. With the protection of privacy information, the image tool by Google doesn't be used to create a map of GPS data otherwise we used Pic2map Photo Location viewer, which has the feature to protect privacy photo by the statement: If you select this option your photos will be hidden from search engines. Otherwise, your photos and location details may be indexed by search engines. If selected, they may also be featured on Pic2Map gallery shown on figure 9. [19]

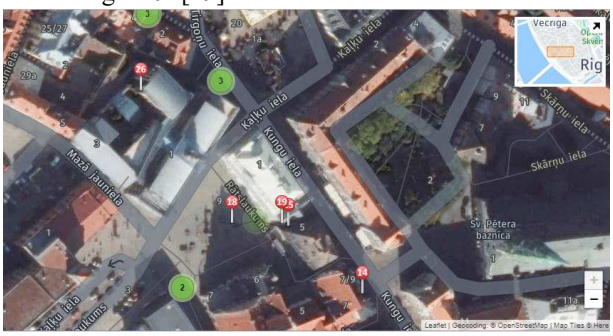

Figure 9. The photo map of Riga

V. Conclusions

In this paper, we present digital computer image forensics in the way of analyzing photos EXIF data. As an example of submission, the set 80 digital photos as digital evidence was used. The system provided visual content of the image, analysis of GPS information, and the application by visualization every image on the map. Our future work will deal with convolutional neural network (CNN) application in the visual content of digital photos with face detection and increase the set of images. At this point would be interested to make a forensics investigation on the fake photos by analyzing the occurrence of the person.

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